Do Now · retrieval

Spiral model and Rapid Application Development

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Name a key XP programming practice.

Q2

How does an Agile sprint help with changed requirements?

Q3

What is a prototype?

Q4

Why might risk analysis be valuable before developing an uncertain hardware-dependent system?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Spiral focuses on risk — Each cycle plans objectives, identifies or resolves major risks, develops/tests and reviews with stakeholders.

Core idea 2

RAD is prototype-led — Users evaluate quickly produced models. Frequent prototyping accelerates understanding and can reveal unclear requirements.

Core idea 3

Compare suitability — Spiral can suit large/high-risk systems but increases management cost; RAD relies on accessible users and skilled teams and may compromise architecture if rushed.

Key visual · complete concept

COMPARE DRIVERS

Spiral	**Manage risk**
RAD	**Rapid feedback**
Waterfall	**Predictable phases**

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

Spiral model — Iterative methodology driven by risk identification and evaluation.

risk analysis — Evaluate probability/impact of potential problems.

RAD — Rapid Application Development using frequent prototypes and feedback.

prototype — Early working or mock representation for evaluation.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

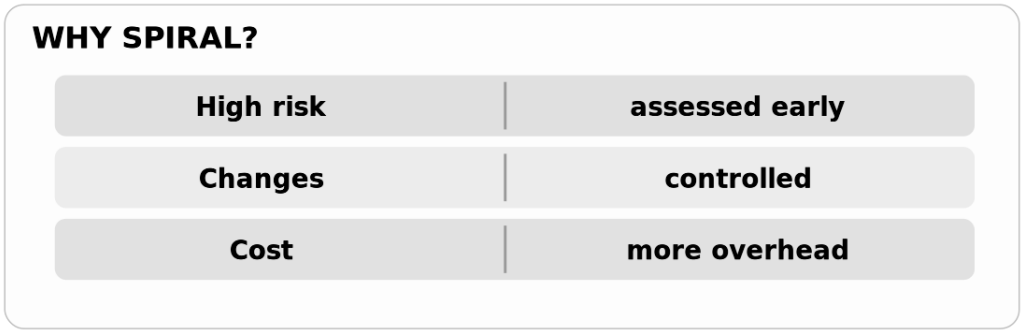
A satellite control project has major safety and technology uncertainties. Recommend a method.

Follow the worked method

1. Predict first — Spiral is appropriate because major technical and safety risks need assessment at each cycle.

2. Model and explain — It demands experienced risk assessment and carries greater time/management overhead than simpler methods.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
RAD suits evolving interface requirements because the café can see and comment on working prototypes.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State one key characteristic and one disadvantage of both Spiral and RAD.

Question 2

A transport company is building an experimental driverless control system. Compare Spiral with RAD and justify a choice.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A booking website must launch rapidly and users are available daily. Explain three benefits and two risks of RAD.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Spiral development and RAD are identical because both can repeat stages.

Use key vocabulary from memory

1. Explain Spiral model in your own words.

2. Explain risk analysis in your own words.

3. Explain RAD in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.